BS24201- Advanced Statistics and Numerical Methods		
Teaching Scheme: **Theory: 3 Hours/Week**	**Credits** **03**	**Examination Scheme:** **Course Activity: 10 Marks** **In-Semester: 30 Marks** **End-Semester: 60 Marks** **Total: 100 Marks**

Prerequisites:

Basics of Statistics, Data Types, Measures of Central Tendency, Probability Theory. Conditional Probability, Bayes Theorem, Basics of Numerical Methods.

Course Objectives:

To make the students familiarize with concepts and techniques in Statistics and Probability Theory. The aim is to equip them with the techniques to understand advanced levels of mathematics, statistics and their applications that would enhance thinking and logical power, useful in their disciplines.

Course Outcomes (COs): The students will be able to learn:

CO1: Understand the concepts of dispersion, skewness and kurtosis, and use them to analyze a given data.

CO2: Understand and apply the theory of correlation and regression analysis for data predictions.

CO3: Classify various discrete probability distributions and apply them to analyze a particular data set.

CO4: Develop basic concepts of continuous probability distributions and use them in their field.

CO5: Learn the concepts of various sampling tests, the null and the alternative hypotheses useful for analyzing the population(s) under study.

CO6: Compute Interpolating polynomials, numerical differentiation and integration, numerical solutions of ordinary differential equations used in modern scientific computing.

Course Contents

Unit I: Measures of Dispersion, Moments, Skewness and Kurtosis (07 Hours)

Measures of Dispersion: Dispersion, Characteristics for an Ideal Measure of Dispersion, Range, Coefficient of Range, Quartile Deviation, Coefficient of Quartile Deviation, Mean Deviation about Mean, Variance, Standard Deviation and Coefficient of Variation.

Moments: Moments about an arbitrary value, Raw moments, Central Moments, Skewness and Kurtosis.

Unit II: Correlation and Regression (07 Hours)

Bivariate Data, Scatter Diagram, and Association between two variables, Correlation, Karl Pearson's coefficient of correlation, Regression: Regression lines, properties, Curve fitting: fitting of straight line, parabola and related curves.

Unit III: Probability and Discrete Probability Distributions (07 Hours)

Probability, Theorems on Probability, Random variables, Discrete Random variable: Probability mass function, Mathematical Expectation, Discrete Probability distributions: Binomial, Poisson, Poisson approximation to Binomial distribution.

Unit IV: Continuous Probability Distributions (07 Hours)

Continuous Random variable: Probability density function, properties, Mathematical Expectation, Continuous Probability distributions: Uniform distribution, Exponential distribution, Gaussian/Normal distribution, Chi-square distribution.

Unit V: Inferential Statistics: Tests For Hypothesis (07 Hours)

Statistical Hypothesis: Simple and Composite, Test of a Statistical Hypothesis, Critical Region, Two Types of Errors, Level of Significance, Power of the Test, Large Sample Tests: Test for Mean of a single Population, Test for the Equality of two Populations Means, Chi-square test for Goodness of fit.

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Unit VI: Numerical Methods (07 Hours) **Interpolation**: Finite Differences, Newton's and Lagrange's Interpolation formula, Numerical Differentiation. **Numerical Integration**: Trapezoidal and Simpson's rules. **Solution of Ordinary differential equations**: Runge-Kutta 4th order method.
Books and Other Resources
Reference Books: 1. Glen Cowan, "Statistical Data Analysis", University Of Siegen, Clarendon Press, Oxford, 1998, ISBN: 0198501552. 2. Ken Black, "Applied Business Statistics", Wiley, 7th Edition, ISBN: 9788126537075 3. Fundamentals of Mathematical Statistics by S. C. Gupta and V. K. Kapoor (Sultan Chand & Sons) 4. Sheldon M. Ross, "Introduction to Probability and Statistics for Engineers and Scientists", 5e, Elsevier Academic Press. 5. M. K. Jain, S. R. K. Iyengar And R. K. Jain1, "Numerical Methods for Scientific and Engineering Computation", 5e, New Age International Publication.
E-Books: 1. NIST/SEMATECH e-Handbook of Statistical Methods 2. Elements of Statistical Learning: data mining, inference, and prediction. 2nd Edition. (su.domains)

Course Coordinator	PAC Member 1	PAC Member 2	Dept. Autonomy Coordinator	BoS Chairman

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